Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-I Business Mathematics Examination May 2018

Introduction:

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part I (Class XI) Business Mathematics. It contains comments on candidates' responses to the 2018 HSSC - I Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E- Marking Notes:

This includes overall comments on candidates' performance on every question and some specific examples of candidates 'responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

Key Observations:

This year candidates faced difficulty in the question based on function and profit & loss. Candidates performed well on questions related to Matrices, Number systems, equation of straight line.

Detailed Comments:

Constructed Response Questions (CRQs)

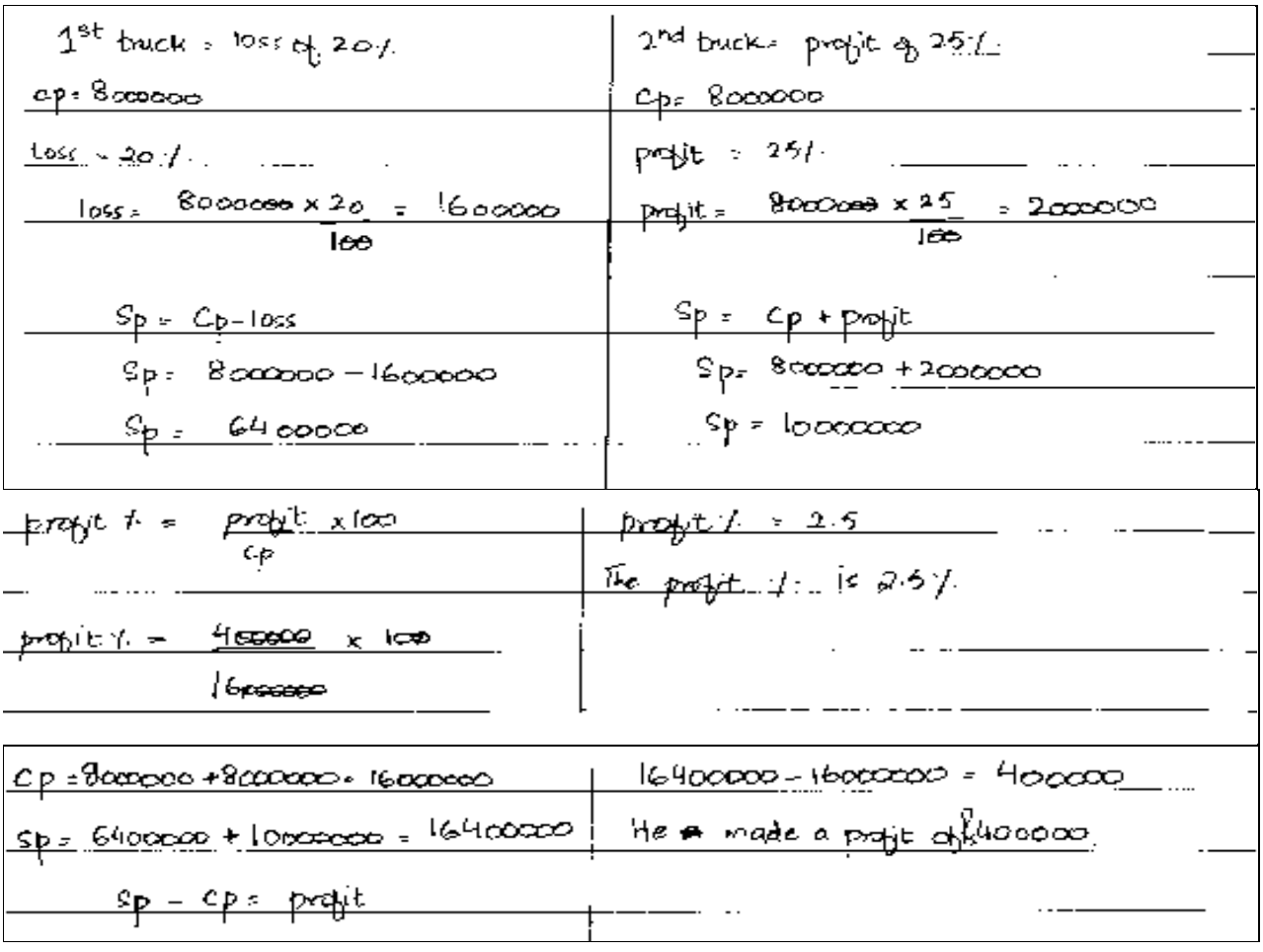
Question 1:

A man bought two trucks for Rs 8,000,000 each. He sells one of these trucks at a loss of 20% and the other truck at a profit of 25%.
Find the

i. selling price of each truck.
ii. total profit he made from selling the two trucks.
iii. percentage profit he made from selling the two trucks.

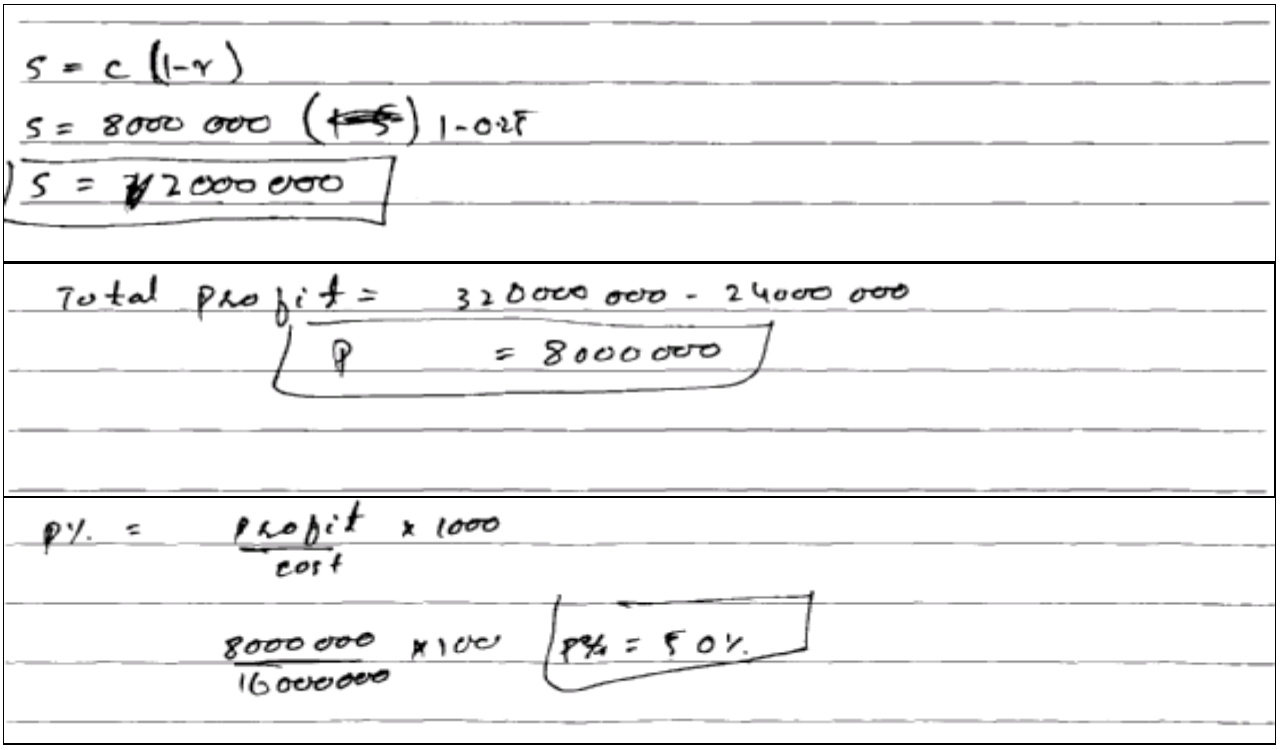
Better responses reported that candidates in part **a** correctly found the selling price of each truck and hence they were able to manage rest of the part quite comfortably. This showed that the candidates had a good knowledge related to profit and loss.

Example:



Weaker responses revealed that candidates lacked understanding about profit and loss. They did not find the selling price of each truck and therefore, they were unable to connect the remaining parts. It was also seen that fewer candidates only managed to calculate 20% of 8,000,000 and 25% of 8,000,000 but they were stuck in proceeding further to get the required answer.

Example:



Question 2:

This question offered a choice between part **a** and part **b**. Majority of the candidates attempted part **a**.

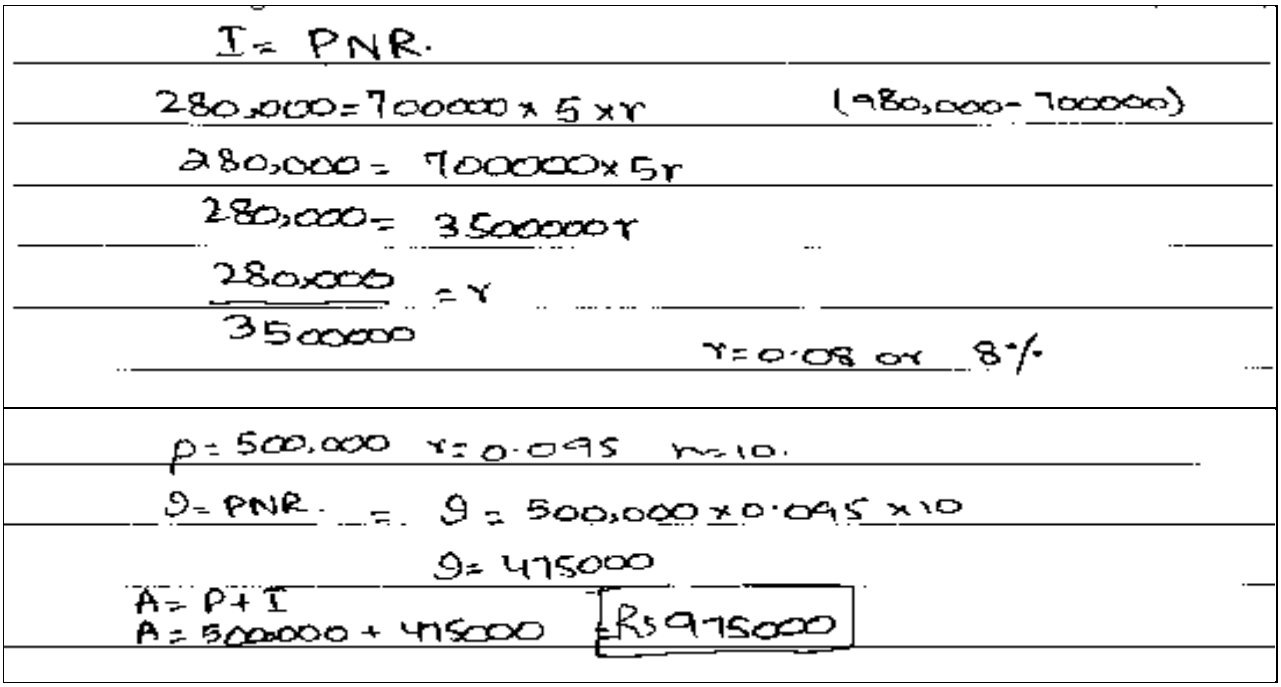
Question 2a:

i. Neha obtained a loan of Rs 700,000 from a financial institute at a simple interest rate of *Z*% per annum. After five years, she made a payment of Rs 980,000 to repay the loan, including the interest. Find the value of Z.

ii. Tabinda deposited Rs 500,000 in a saving scheme at a simple interest rate of 9.5% per annum. Find the accumulated amount Tabinda will get after 10 years.

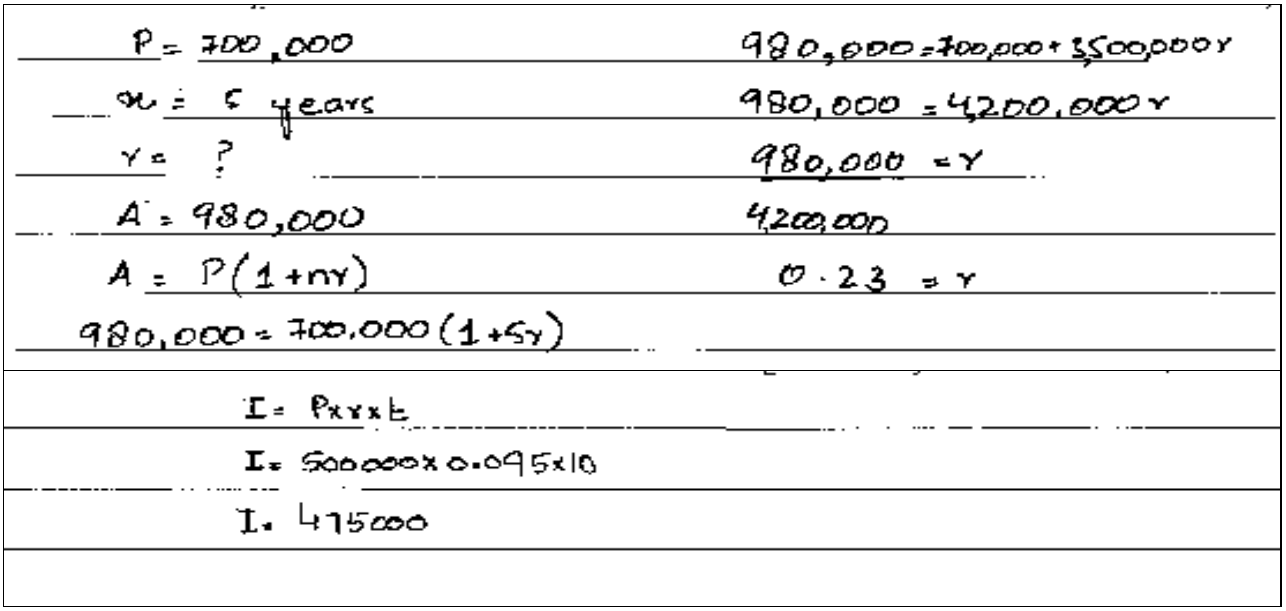
Better responses informed that the candidates in both the parts applied their knowledge to calculate the interest rate based on the concept of simple interest. It was noticed that candidates were able to comprehend the question well and used correct formula to get the required answer. These responses were free from calculation errors as well which highlighted the candidates' ability of attempting the question.

Example:



Weaker responses indicated that candidates were unable to differentiate between formula of simple interest and compound interest. Some weaker responses reported that candidates failed to find the required interest rate and the accumulated amount after 10 years because they made calculation errors.

Example:

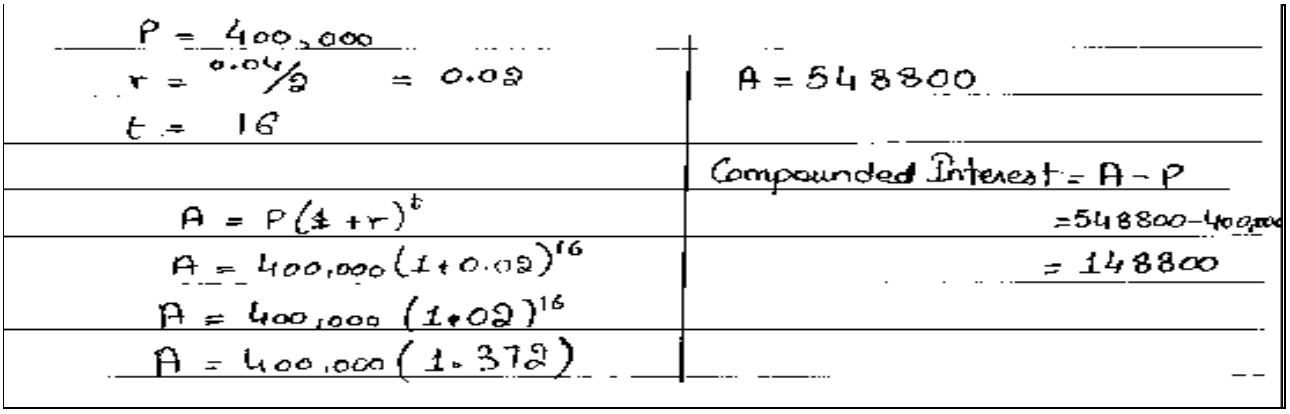


Question 2b:

Saad deposits Rs 400,000 in his savings account at a bank. The account earns 4% interest per year, compounded semi-annually. What will be the accumulated amount in his account after 8 years?

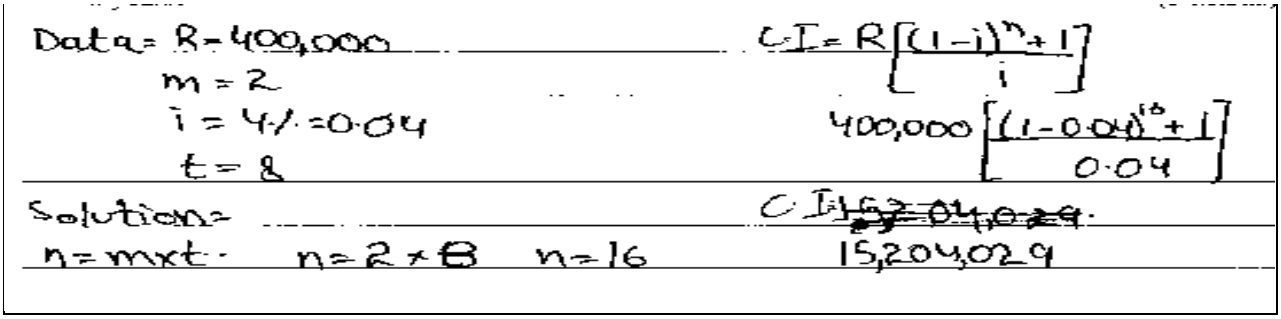
Better responses revealed that candidates systematically determined the values of *i* and number of compounding periods *n*. This helped them in applying the formula of compound interest correctly to get the quantities required in the question.

Example:



Weaker responses indicated that candidates were unable to differentiate between principal amount and accumulated amount and therefore, failed to apply the formula correctly. Some candidates also made calculation errors while finding compounding periods and interest rate.

Example:



Question 3:

This question offered a choice between part **a** and part **b**. Both the parts were attempted equally.

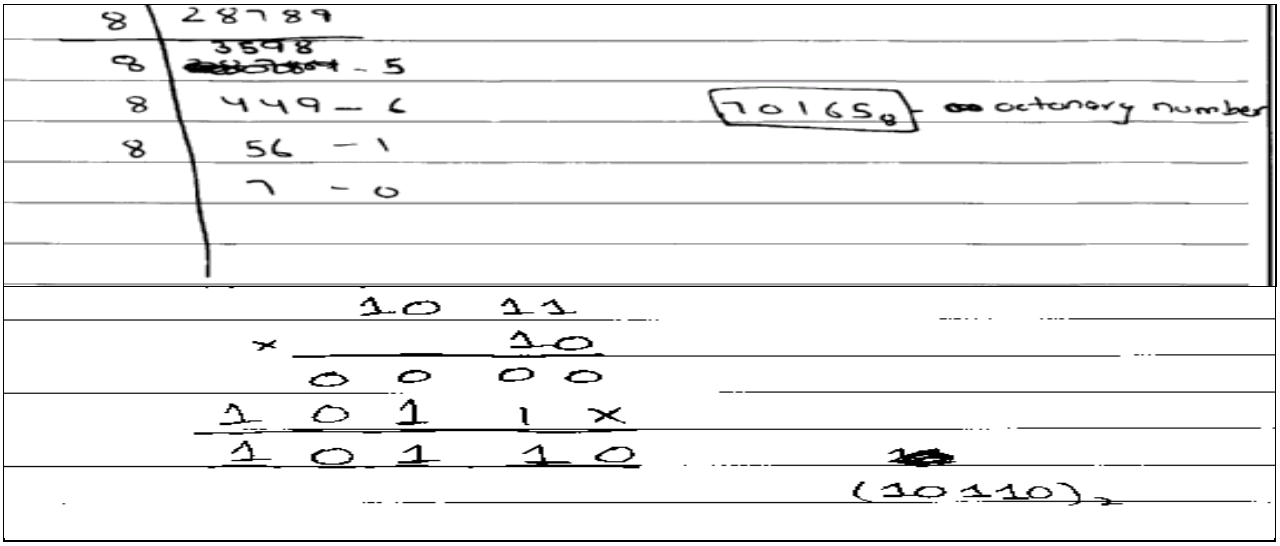
Question 3a:

i. Convert the decimal number 28,789 into an octonary number. Show your working.

ii. Multiply the binary numbers 1011 and 10. Show your working.

Better responses reported that the candidates in part i comfortably converted the given number in base 10 to number in base 8. Similarly, in part ii, they multiplied the given binary numbers quite easily. These responses by the candidates demonstrated a clear understanding of the concept being examined.

Example:



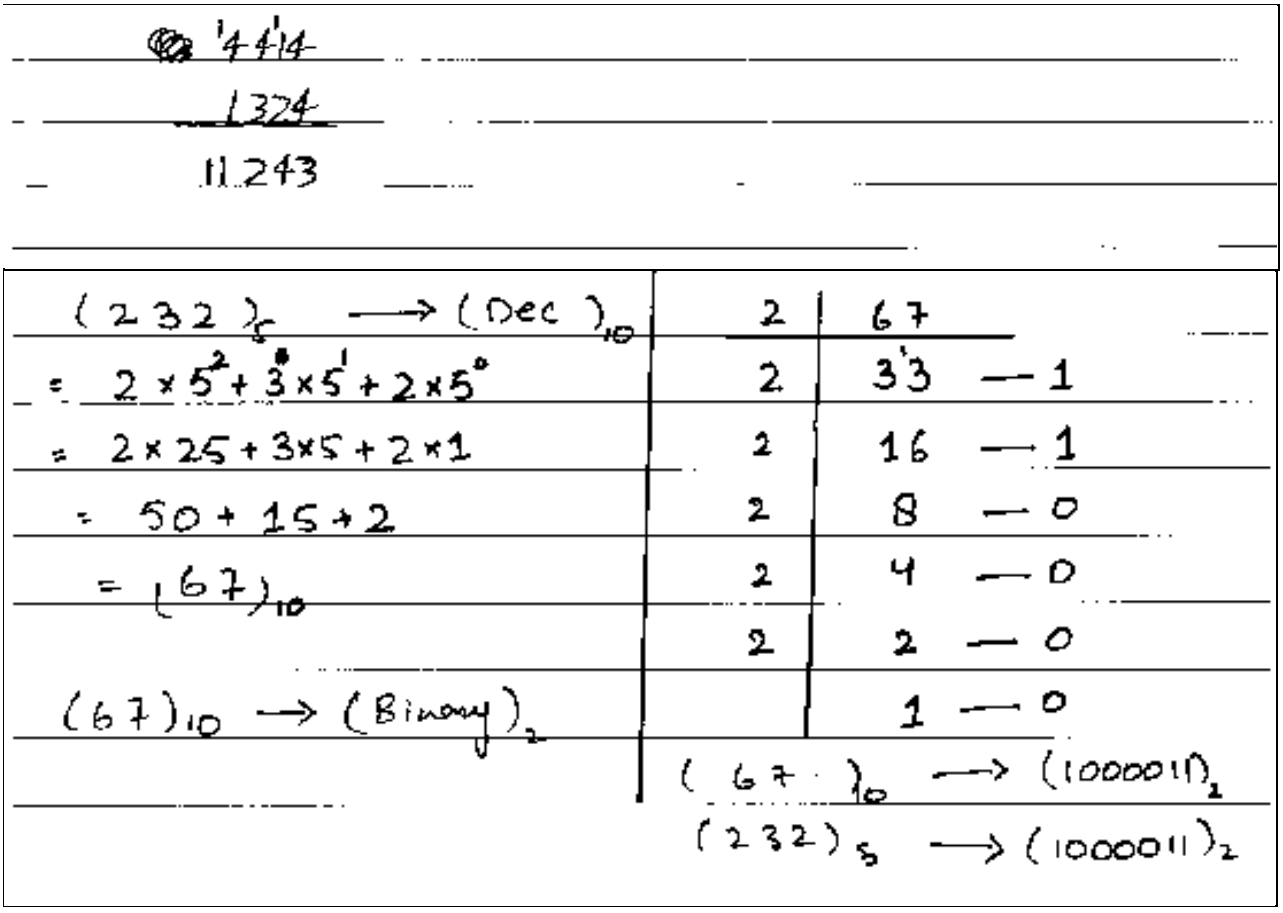
Question 3b:

i. Add $(1324)_5$ to $(4414)_5$.

ii. Convert $(232)_5$ into a binary number.

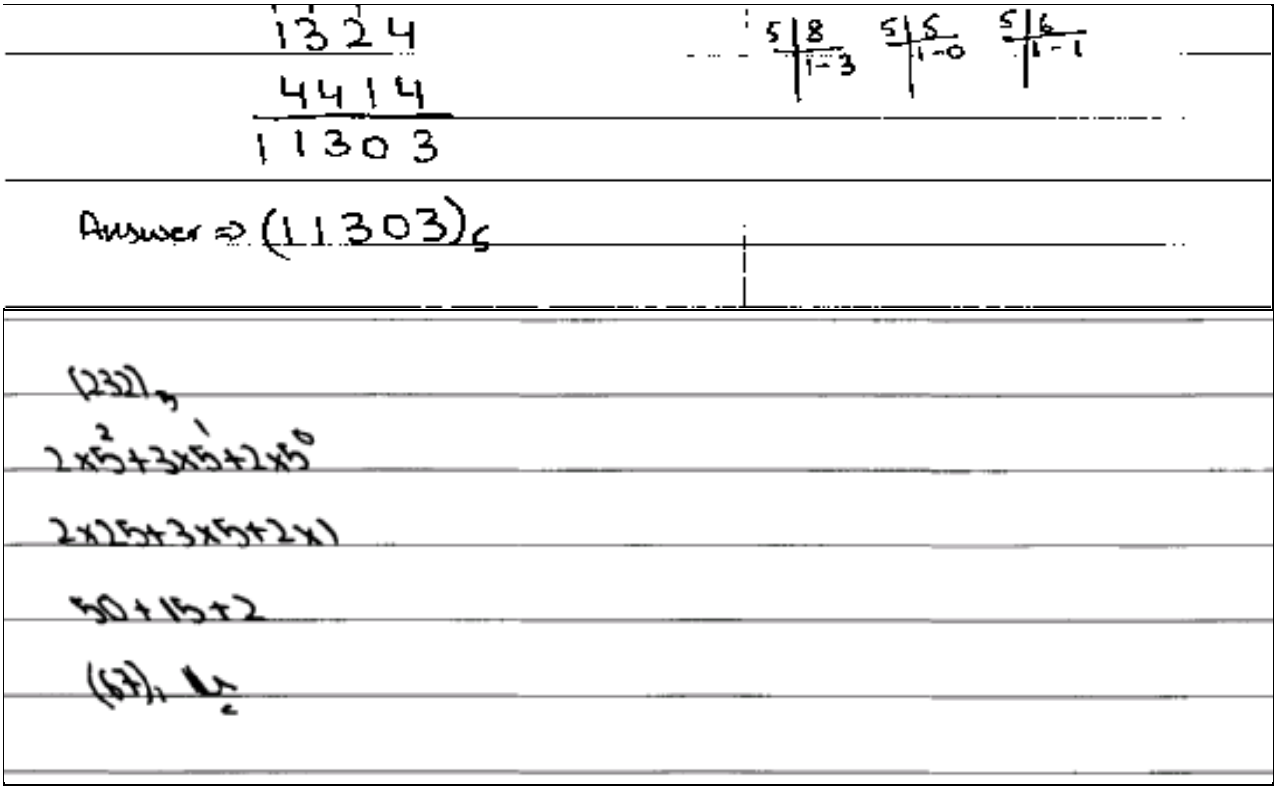
Better responses indicated that the candidates in part i were able to understand and apply their skills in the operations of addition and multiplication on different number systems. Similarly, in part ii, they converted all the numbers to their decimal equivalents and then, performed the required operation skilfully to get the answer.

Example:



Weaker responses indicated that the candidates failed to perform the number operations on the given numbers correctly. They mainly failed to understand the importance of conversion of each number to same base or base 10. Secondly, they incorrectly converted number from different bases to base 10. It was also noticed that fewer candidates just managed to convert from base five to decimal number system but after that they got stuck to proceed further which led to an incomplete and incorrect answer.

Example:



Question 4:

This question offered a choice between part **a** and part **b**. Candidates performed well in part **b** but both parts were almost equally attempted by the candidates.

Question 4a:

i. For a set $A = \{10, 20, 30\}$, determine whether the following are a binary relation or a function from A to A.

   I. $\{(10, 10), (10, 20), (10, 30)\}$

   II. $\{(10, 20), (20, 10), (30, 30)\}$

ii. For the parabola $y = (x-3)^2 + 1$, find the

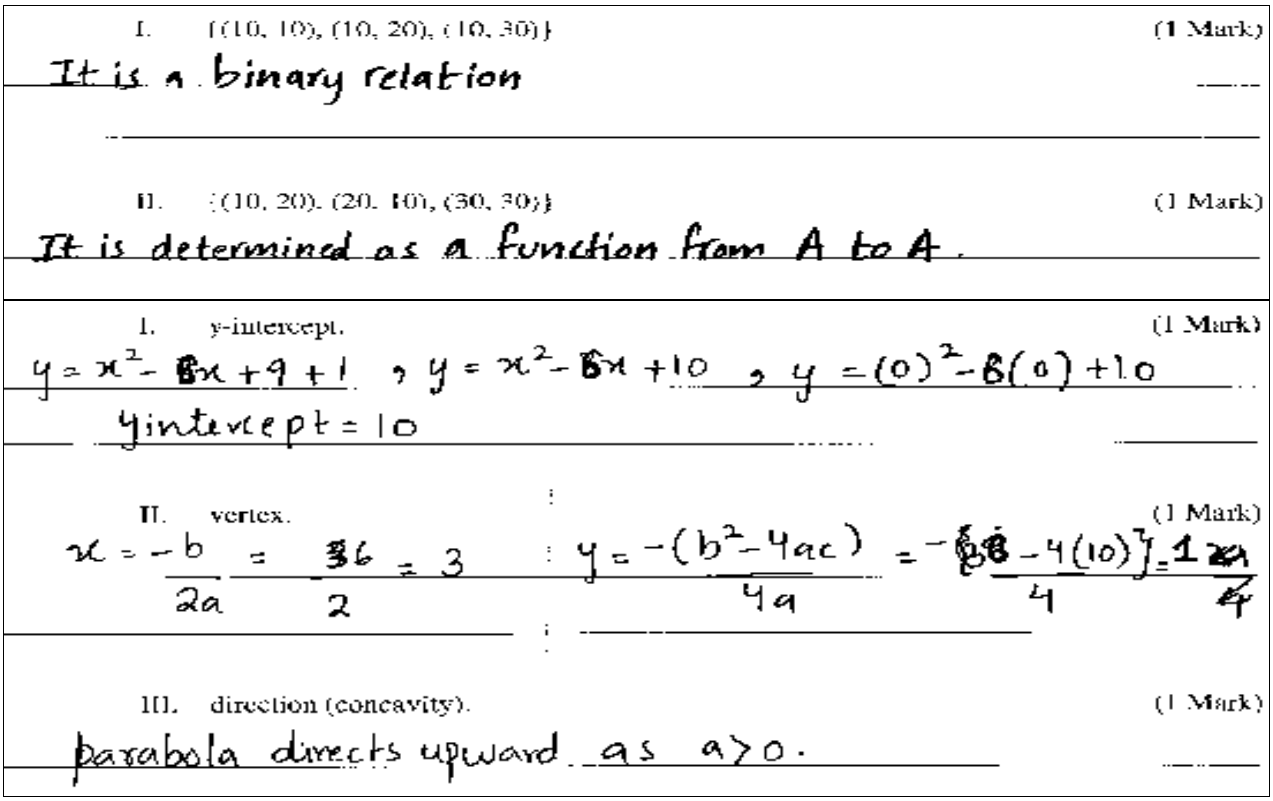
   I. y - intercept.

   II. vertex.

   III. direction (concavity).

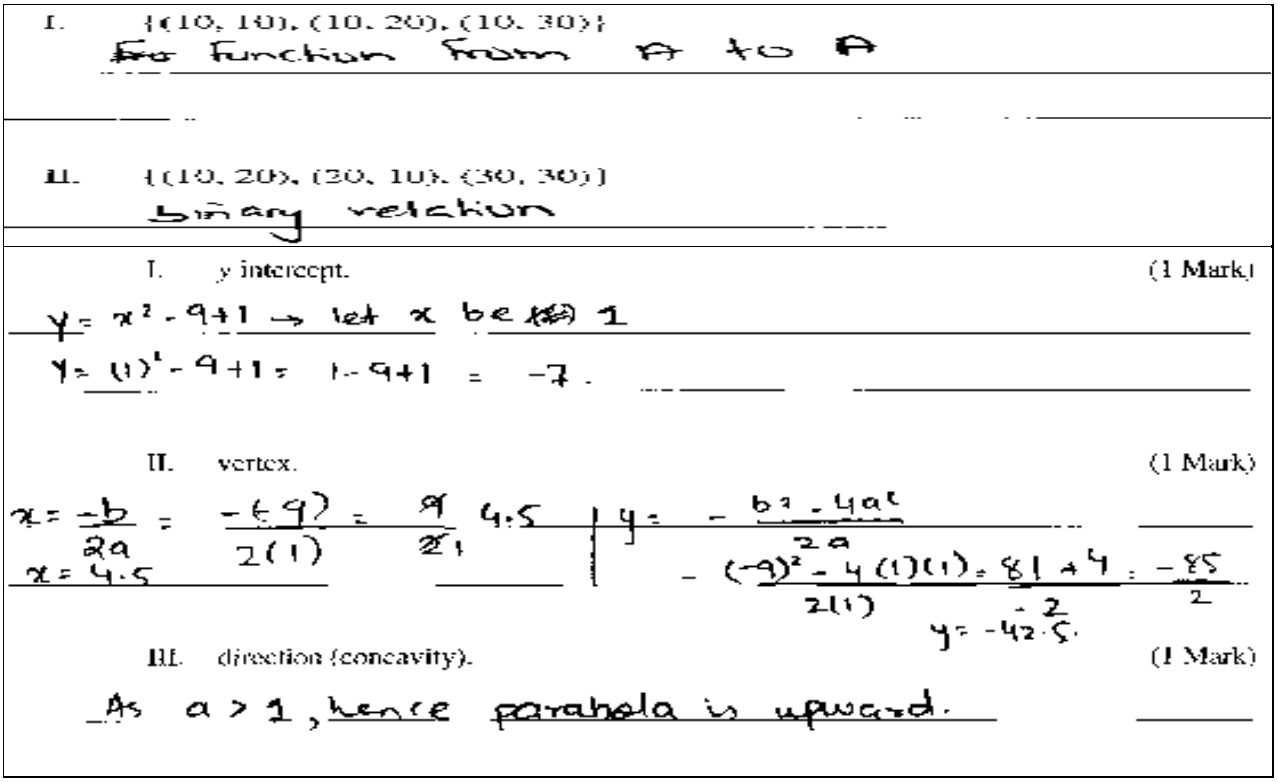
Better responses indicated that candidates in part i had a good understanding of concepts based on functions. This helped candidates to differentiate between binary relation and function quite easily. As a result, they got maximum marks. Similarly, in part ii, it was observed that candidates used the equation of parabola properly so that they were able to determine y – intercept and vertex. Because of this clarity in concepts, they were comfortable to mention the direction (concavity) as well.

Example:



Weaker responses displayed that in both the parts candidates failed to grasp the concepts related to functions and parabola. In part i, some candidates were totally blank whilst in part ii, they could not use the equation properly and they were unable to proceed further which resulted loss of marks.

Example:



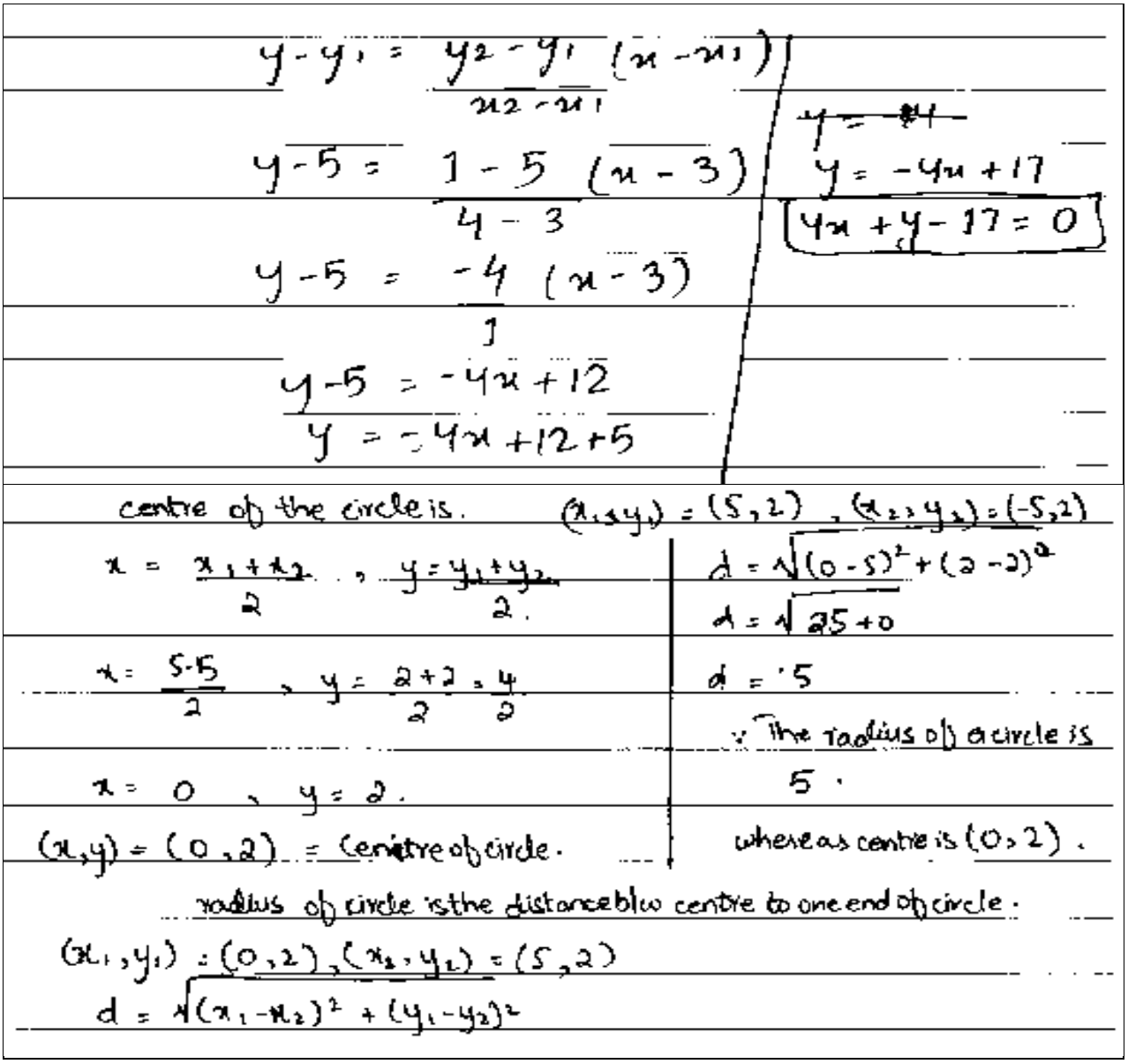
Question 4b:

i. Find the equation of a line that passes through the points (3, 5) and (4, 1). Express your answer in the form .

ii. The end points of the diameter of a circle lie on the points (5, 2) and (– 5, 2). Find the centre and radius of the circle.

Better responses in part i selected the correct form of the equation and applied the form aptly to find the required equation. Also in part ii, they were coherently used the points in order to calculate radius and centre of the circle.

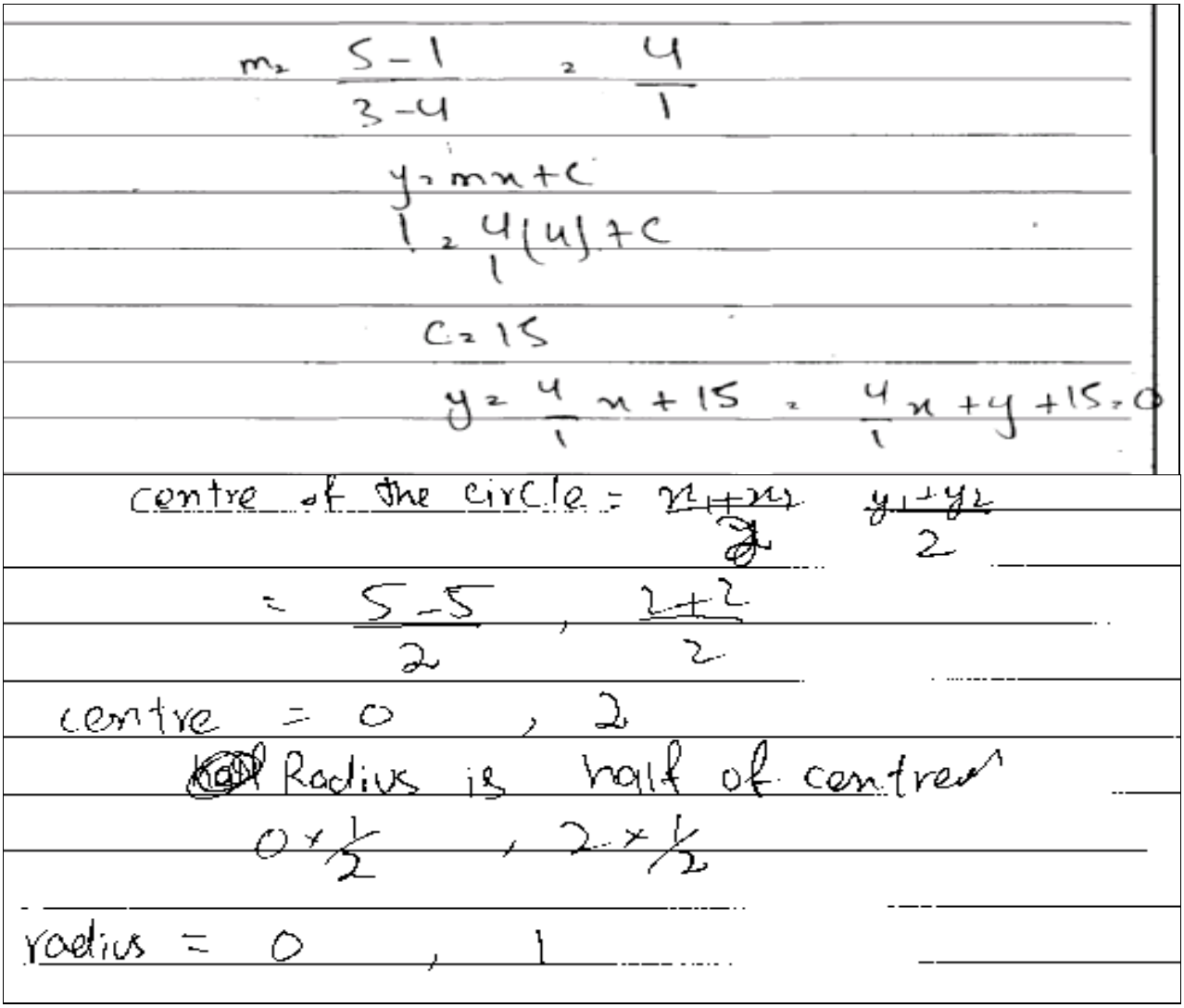
Example:



Weaker responses displayed that in part i, candidates did not understand the question. They were confused among the different forms of straight line.

Similarly in part ii, candidates although used the formula correctly but made mistakes in substituting the values which led to an incorrect answer. It was also seen that candidates failed to calculate the radius because of wrong selection of coordinates. They used (0, 0) and (0, 2) rather than (5, 2) and (– 5, 2) or (0, 2) and (– 5, 2) or (5, 2) and (0, 2) which resulted the loss of marks.

Example:



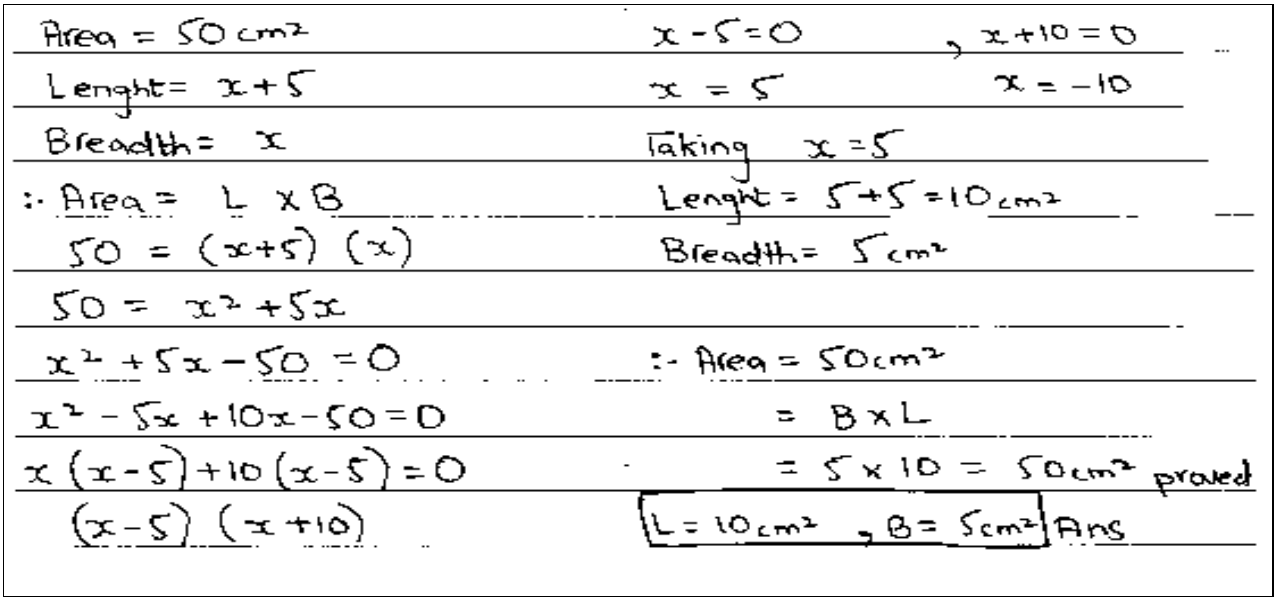
Question 5:

The area of a rectangle is 50 cm^2. Its length is 5 cm more than its breadth. Find the length and breadth of the rectangle.

This was generally a well attempted question.

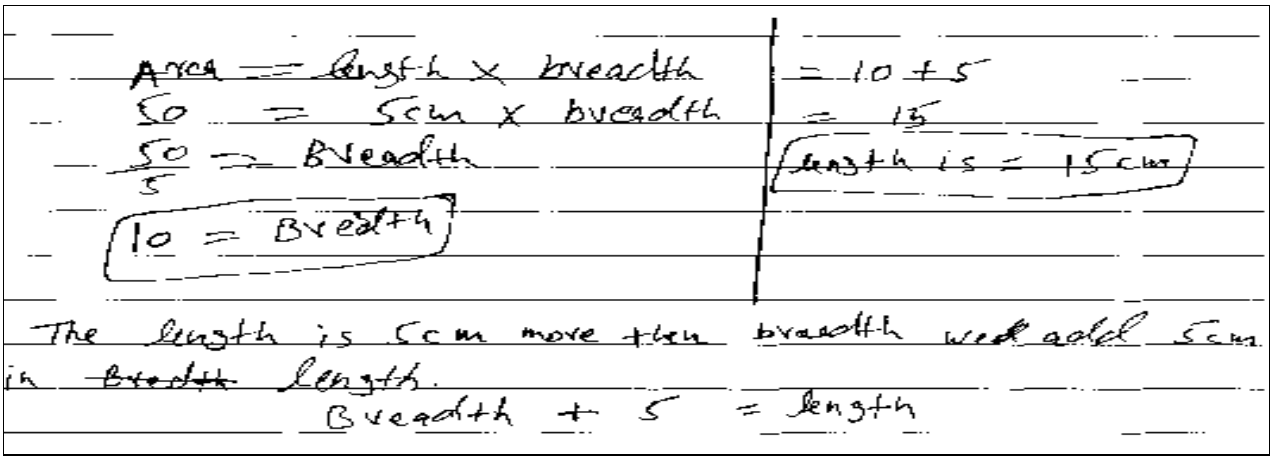
Better responses reported that candidates were able to translate the given situation mathematically. This assisted candidates to find the length and breadth of a rectangle aptly.

Example:



Weaker responses showed that the candidates were unable to translate and used the given situation accordingly. It was also noted that fewer candidates chose larger value of breadth instead of length. In some cases, though they wrote the equation correctly but, could not factorize to get the desired answer which indicated the lack of practice.

Example:



Question 6:

a. Find the values of p and q from the given matrix equation.

$$3\begin{bmatrix} 2 & p \\ 1 & 5 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ q & 15 \end{bmatrix}$$

b. Without using calculator, find $\begin{bmatrix} 1 & 3 & 2 \\ 5 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 5 \\ 1 & 2 \\ 5 & 0 \end{bmatrix}$. Also state the reason why these two matrices are conformable for multiplication.

Better responses demonstrated that, in part **a**, candidates were well-versed in solving the problem based on the concept of matrices operations. Candidates did well in scalar multiplication followed by the addition of matrices and finally compared the matrix $\begin{bmatrix} 7 & 3p + 2 \\ 6 & 15 \end{bmatrix}$ with $\begin{bmatrix} 7 & 2 \\ q & 15 \end{bmatrix}$. As a result, they found the required values.

In part **b**, candidates were well aware of multiplication of two matrices and because of this, they were able to write the correct reason for the multiplication of two matrices.

Example:

$$\begin{bmatrix} 6 & 3p \\ 3 & 15 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ q & 15 \end{bmatrix}$$

$$\begin{bmatrix} 6+1 & 3p+2 \\ 3+3 & 15+0 \end{bmatrix} \quad \begin{bmatrix} 7 & 2 \\ q & 15 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 2 \\ 6 & 15 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 6 & 15 \end{bmatrix}$$

$p = 6$

$q = 6$

$3p+2 = 2$

$3p = 2-2$

$3p = 0$

$p = \frac{0}{3}$

$p = 0$

$$\begin{bmatrix} 1 & 3 & 2 \\ 5 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 5 \\ 1 & 2 \\ 5 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0+3+10 & 1\times5+3\times2+2\times0 \\ 5\times0+1\times1+0\times5 & 5\times5+1\times2+0\times0 \end{bmatrix} = \begin{bmatrix} 3+10 & 5+6+0 \\ 0+1+0 & 25+2+0 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & 11 \\ 1 & 27 \end{bmatrix}$$

These two matrices are comfortable to multiply because Colomn of first matrix is equal to row of second matrix. for multiplication it is important that colomn of 1st matrix should be equal to row of 2nd Matrix. and in this case it is equal. 3 colomns = 3 rows.

Weaker responses displayed that the candidates failed to attempt both the questions because of vague understanding and lack of practice as well. It was noted that candidates were perplexed in addition and multiplication of two matrices. In part i, they made mistakes in comparing the corresponding entries which resulted the loss of marks. Similarly in part ii, they could not multiply the given matrices and hence were unable to give the right reason of multiplication of two matrices.

Example:

